

THE CALABI–YAU-TO-CHIRAL FUNCTOR Φ AND THE UNIVERSAL TRACE IDENTITY

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ABSTRACT. The Calabi–Yau-to-chiral functor $\Phi: \text{CY}_d \rightarrow \text{Chiral-Alg}$ assigns to a compact Calabi–Yau d -fold X a chiral factorisation algebra $\mathcal{A}_X$ on an auxiliary curve. Its principal invariant is $\kappa_{\text{ch}}(\mathcal{A}_X) = \sum_q (-1)^q h^{\circ, q}(X)$ (Hodge supertrace, proved with separate algebraic and geometric models). The output E_n -scope stratifies by dimension: $\Phi(X) \in E_2\text{-chiral}$ for $d \leq 2$ and $\Phi(X) \in E_1\text{-chiral}$ for $d \geq 3$. The universal trace identity unifying Vol I $K = -c_{\text{ghost}}(\text{BRST})$ with $\kappa_{\text{BKM}}(\Phi_N) = c_N(o)/2$ connects Vol I’s universal conductor with Vol III’s Borchers weight.

SCOPE AND STATUS

Primary theorems:

- Φ preserves Koszul duality on the Koszul locus: .
- $\kappa_{\text{ch}} = \chi_{\text{Hodge}}$ (Hodge supertrace): (thm:kappa-hodge-supertrace-identification).
- E_n -output scope $n = 2$ for $d \leq 2$, $n = 1$ for $d \geq 3$: (FM43 scope guard).
- Universal trace identity $K_{\text{univ}} = -c_{\text{ghost}}$: (Vol I chap:universal-conductor).

CHAPTER INPUTS

The standalone records the following chapter sources:

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\input{../chapters/theory/cy_to_chiral.tex}
\input{../chapters/theory/quantum_chiral_algebras.tex}
\input{../chapters/theory/en_factorization.tex}
\input{../chapters/examples/cy_d_kappa_stratification.tex}
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CROSS-VOLUME DEPENDENCIES

- Vol I chap:universal-conductor (universal trace identity $K_{\text{univ}} = -c_{\text{ghost}}$).
- Vol I thm:A-infinity-2 (properad bar-cobar; Vol III Φ is a corollary of Francis–Gaitsgory (cor:chiral-KK-formal-smoothness)).
- Vol II thm:chiral-higher-deligne (target category for $\Phi(X)$ ’s derived centre at $d = 3$).